



CSE422: ARTIFICIAL INTELLIGENCE

PROJECT REPORT

SUMMER 2026

Adult Income Prediction Using Machine Learning

SECTION 01 – GROUP 04

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TABLE OF CONTENTS

CSE422: ARTIFICIAL INTELLIGENCE.....	0
PROJECT REPORT.....	0
SUMMER 2026.....	0
Adult Income Prediction Using Machine Learning.....	0
INTRODUCTION.....	2
PROJECT OVERVIEW.....	2
DATASET DESCRIPTION.....	3
DATA ANALYSIS.....	3
DATA PRE-PROCESSING.....	12
DATASET SPLITTING.....	13
SUPERVISED MODEL TRAINING AND EVALUATION.....	14
1. NEURAL NETWORK.....	14
2. LOGISTIC REGRESSION.....	16
3. DECISION TREE.....	17
COMPARATIVE ANALYSIS OF MODELS.....	19
K-FOLD CROSS VALIDATION.....	20
UNSUPERVISED MODEL TRAINING AND EVALUATION.....	22
K-MEANS CLUSTERING.....	22
CONCLUSION.....	25

INTRODUCTION

In this project, different machine learning techniques have been applied to the given dataset on 'adult income', with the objective of predicting an individual's annual income category based on their demographic, educational and occupational attributes. The income prediction task is treated as a binary classification problem, categorizing individuals by the income of either $\leq 50K$ or $> 50K$. In order to examine the effectiveness of the different techniques, models such as Neural Network, Logistic Regression, and Decision Tree have been applied and compared using standard evaluation metrics. Additionally, K-Means clustering is used to identify the hidden patterns and groups within the dataset.

PROJECT OVERVIEW

This project focuses on the implementation of different machine learning techniques to the given dataset on '**adult income**' in order to analyze and predict an individual's annual income based on the collected demographic data, educational, occupational and employment-related characteristics. The primary goal is to treat the income prediction as a **binary classification** problem, where the target variable represents whether the annual income of an individual is classified as $\leq 50K$ or $> 50K$.

This project includes three different supervised learning approaches, including **Neural Network, Logistic Regression, and Decision Tree**, to determine how different models classify the individuals into the two income categories. All models are evaluated using **accuracy, precision, recall, F1 score, confusion matrices, ROC curve and AUC score**, along with **K-Fold Cross Validation** to assess the performances across different subsets of the dataset.

In addition to supervised learning, this project implements an unsupervised learning approach using K-means clustering. This allows the identification of naturally forming groups or clusters within the dataset without using the target variable during the process of cluster formation. The results are then interpreted and analyzed to understand the patterns within different clusters.

Overall, the core aim of the project is to compare different machine learning models, identify the most effective model for the income category prediction, and finally investigate the patterns within the given dataset.

DATASET DESCRIPTION

The '**Adult Income**' dataset contains demographic, educational and occupational information about individuals, with the aim to determine their income category. It contains **48,842 records and 14 input features**

including the **target feature** called ‘target’. The target represents two distinct income categories, **<=50K and >50K**, which were mapped to binary values **0 and 1**, respectively, to carry out the machine learning process. After data cleaning, preprocessing, and feature selection, the final dataset contained **48,790 records and 12 input features**, excluding the target.

The input features can be divided into **quantitative and categorical features**. The quantitative features represent numerical or count-based characteristics of an individual, while the categorical features represent attributes consisting of groups or categories. The **quantitative** features are *Age*, *Final Weight*, *Education Number of Years*, *Capital Gain*, *Capital Loss*, and *Hours per Week*, while the **categorical** features are *Workclass*, *Education*, *Marital Status*, *Occupation*, *Relationship*, *Race*, *Sex*, and *Native Country*, as well as the *target* variable.

The ‘target’ feature has been selected as the target variable because the main goal of the project is to predict an individual’s income category based on the given attributes. As it consists of two distinct categories, the problem is concluded as a **binary classification problem**. The dataset, therefore, provides a suitable basis for applying multiple supervised classification models like **Neural Network, Logistic Regression and Decision Tree**, as well as an unsupervised learning technique that includes **K-Means Clustering**.

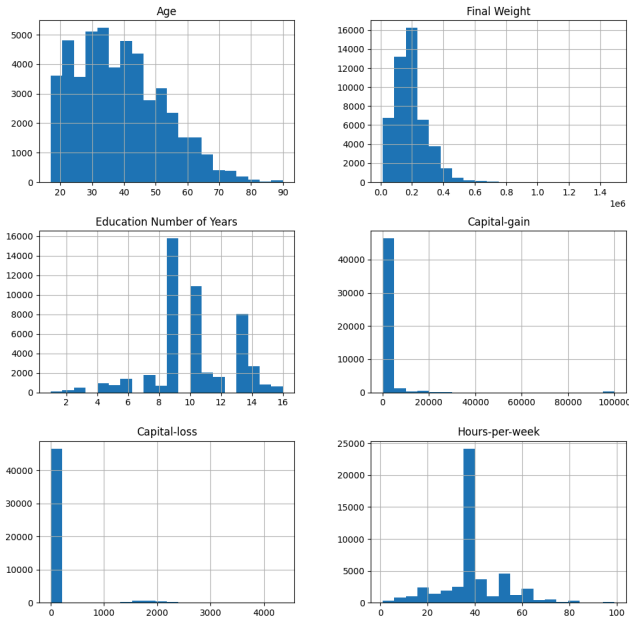
DATA ANALYSIS

Numerical Features

Data Distribution:

The histograms show the distribution of the six numerical features in the Adult Income dataset. *Age* and *Hours-per-week* show relatively concentrated distribution, while *Final Weight*, *Capital-gain*, and *Capital-loss* are strongly right-skewed. *Education Number of Years* is concentrated around higher education levels, particularly around 9-13 years.

The distributions show that several numerical features contain significant skewness, especially *Capital-gain* and *Capital-loss* have most observations concentrated near zero with a small number of very large values. *Final Weight* is also heavily right-skewed. These differences in scale and distribution justify the use of **feature scaling during data preprocessing**, while the extreme values should not automatically be removed because they may be valid observations.

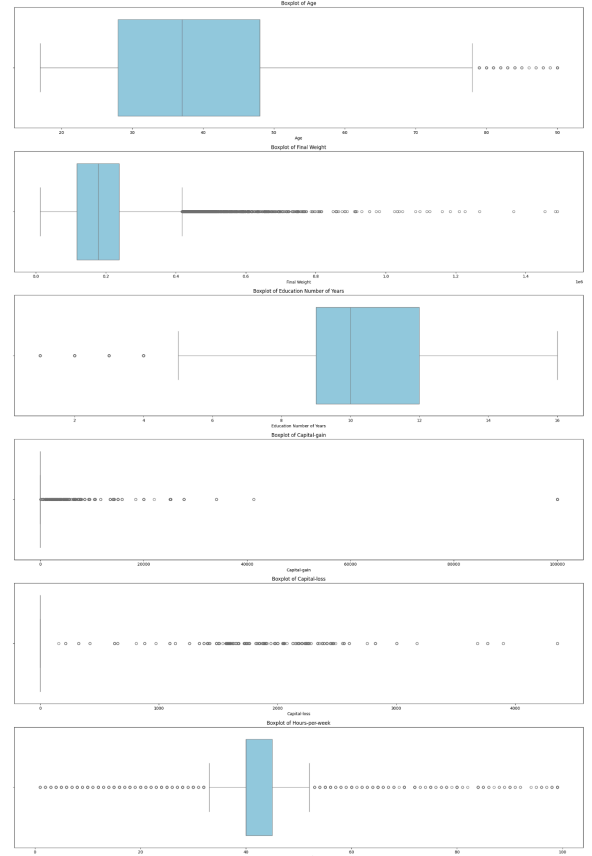


Outlier Analysis:

The box plots show the spread and existing outliers of the numerical features. *Age* and *Education Number of Years* contain relatively limited extreme observations, whereas *Final Weight*, *Capital-gain*, and *Capital-loss* contain numerous points beyond their upper whiskers. *Hours-per-week* also shows observations at both lower and higher ranges.

The presence of outliers is particularly noticeable in *Capital-gain*, *Capital-loss*, and *Final Weight*. These values contribute to the strong skewness observed in the histograms. Since these extreme observations may represent actual individuals rather than errors, they were kept rather than being removed. Instead, **standardization** was applied during

preprocessing to reduce the influence of differences in numerical values.



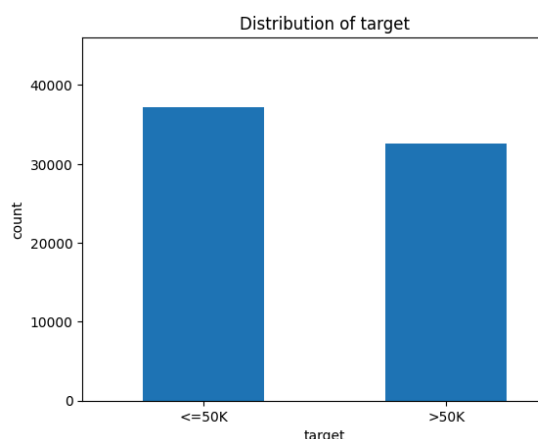
Categorical Features

Target Distribution:

The target distribution shows the number of individuals belonging to the two income categories, $\leq 50K$ and $>50K$. The $\leq 50K$ category represents the majority of observations, while the $>50K$ category represents a smaller portion of the dataset.

The distribution indicates that the target variable is imbalanced, with a larger percentage of individuals earning $\leq 50K$ and a smaller portion earning $>50K$. This imbalance is

important during model development because a model could achieve high accuracy by favoring the majority class. Therefore, accuracy was not considered sufficient on its own, and additional metrics such as precision, recall, F1-score, and AUC were used for evaluation.

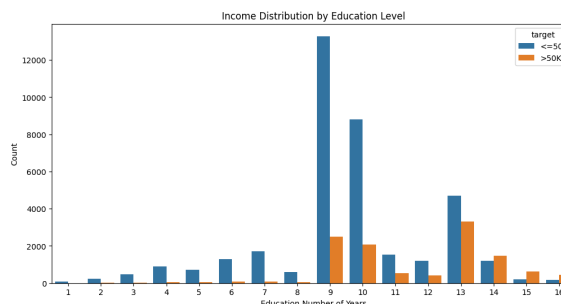


Feature-Target Analysis

The chart compares the number of individuals in the two income categories across different values of Education Number of Years. The distribution shows that lower education levels contain significantly more individuals in the <=50K category, while the proportion of >50K individuals generally increases as education levels increase.

The graph suggests a clear relationship between obtaining education and income category. Individuals with higher numbers of education years are more frequently

represented in the >50K category, particularly at the upper education levels. This observation is consistent with the correlation analysis, where Education Number of Years has the strongest positive relationship with the target among the numerical features.

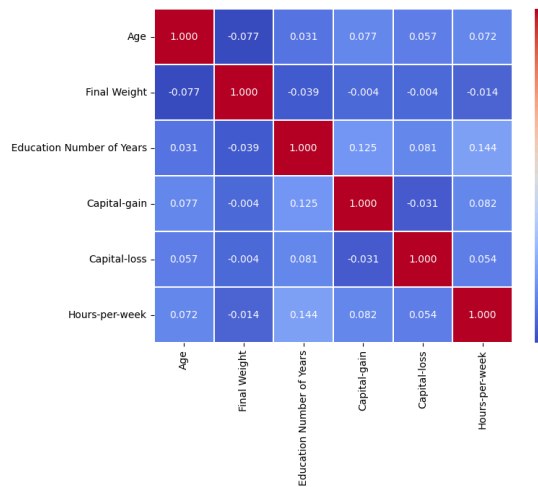


Correlation Analysis

Feature-Feature Correlation Heatmap:

The correlation heatmap illustrates the correlation coefficients among the numerical features. Most features have weak correlations with each other, meaning they do not strongly depend on one another. Therefore, there is no major issue of duplicated information among the numerical features.

The numerical features have mostly weak relationships with each other. For example, Education Number of Years has only a small relationship with Hours-per-week and Capital-gain. Final Weight also has very little relationship with the other features. This means there is no major problem of features giving the same information.

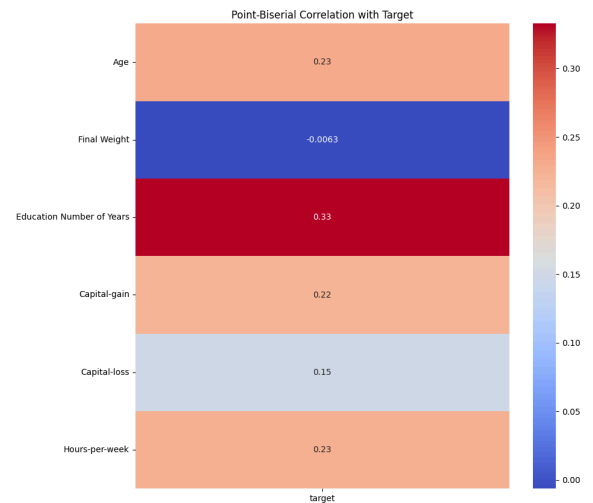


Point-Biserial Target Heatmap:

The point-biserial correlation heatmap shows the relationship between each numerical feature and the binary income target. Education Number of Years has the strongest positive correlation with the target at approximately 0.33, followed by Age and Hours-per-week at approximately 0.23, and Capital-gain at approximately 0.22.

The results indicate that Education Number of Years is the most strongly associated numerical feature with income category. Age, Hours-per-week, and Capital-gain also show moderately positive relations, suggesting that higher values of these features tend to be associated with the >50K category. Capital-loss has a weaker positive relation, while Final Weight has an almost negligible correlation of approximately -0.0063. These findings supported the **feature**

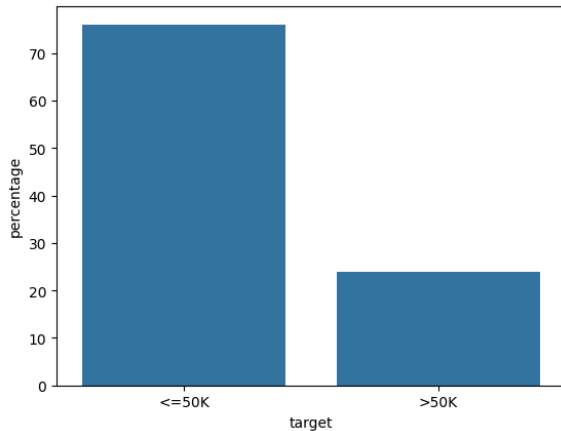
selection process, particularly the decision to remove Final Weight.



Class Distribution

The class distribution shows that the <=50K income category constitutes approximately 76% of the dataset, whereas the >50K category accounts for approximately 24%.

This shows a noticeable class imbalance in the target variable. The minority >50K class is substantially smaller than the majority class, which can make accuracy alone misleading when evaluating classification performance. Therefore, precision, recall, F1-score, confusion matrices, and ROC-AUC were also considered during model evaluation.



DATA PRE-PROCESSING

Duplicate Removal

Duplicate records were examined and removed to ensure that repeated observations would not introduce bias into the model during the training process. A total of 52 duplicate records were identified and removed from the dataset.

Missing Value Handling

Missing values were examined and removed across all features. Even though the dataset initially contained no 'null' values, several categorical attributes contained records with '?' indicating missing values. These placeholders were converted to null values so that they could be easily handled during data preprocessing. This, therefore, was done by replacing the null values with most frequently occurring values in the corresponding feature.

Feature Selection

Feature selection was performed in order to remove redundant and less informative attributes before the training process. The 'Education' feature was removed as it simply provides categorical representation of educational experience which has its own numerical feature called 'Education Number of Years'. Having both information would introduce duplicates in the model.

'Final Weight' was removed because it is a sampling-related attribute and was not directly useful for prediction.

Encoding and Scaling

Since the dataset contains both numerical and categorical features, separate preprocessing techniques were applied to each feature type. Numerical features were standardized using 'StandardScaler' so that features with larger numerical ranges would not disproportionately influence the models. Categorical features were converted into numerical values using 'One-Hot Encoding', where each category is represented by a binary value.

One-Hot encoding was selected as the categorical attributes do not have a meaningful numerical ordering. This prevents the model from incorrectly interpreting categories as having a ranked relationship.

A 'ColumnTransformer' was used to implement the appropriate preprocessing operations to numerical and categorical features separately. The preprocessing pipeline was fitted only on the training data and subsequently applied to the test data to

prevent information from the test set from influencing model training.

DATASET SPLITTING

Since the target variable is **imbalanced**, with approximately 76% of samples belonging to $\leq 50K$ and 24% to $> 50K$, we used **stratified** sampling to preserve this class distribution in both the training and testing sets.

The dataset was divided into:

- **Training set: 80%** $\rightarrow$ 39,032 samples
- **Testing set: 20%** $\rightarrow$ 9,758 samples

A '*random_state*' of 42 was used to ensure reproducibility. Stratification was performed using the target variable so that both subsets, training and testing, maintained approximately the same proportion of the two income categories.

The training set was used for model training and cross-validation, while the test set was kept separate for the final evaluation of the supervised models. This ensures that the final performance metrics reflect how the trained models perform on previously unseen data.

SUPERVISED MODEL TRAINING AND EVALUATION

Three supervised machine learning models were used to predict whether an individual's income is $\leq 50K$ or $> 50K$: Neural Network, Logistic Regression, and Decision Tree. Each model was trained using the preprocessed training data and evaluated on the test data using accuracy, precision, recall, F1-score, confusion matrix, and ROC-AUC. These metrics were used to understand how well each model performed.

1. NEURAL NETWORK

A Neural Network was implemented using three dense layers. ReLU activation was used in the hidden layers, while a sigmoid activation function was used in the output layer for binary classification. The model was trained using the Adam optimizer and binary cross-entropy loss.

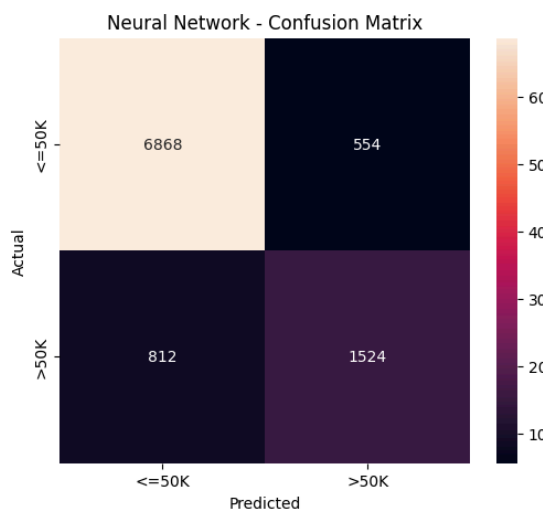
Evaluation Results:

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Neural Network Performance
-----
Accuracy:  0.8600
Precision:  0.7334
Recall:     0.6524
F1 Score:   0.6905
```

The Neural Network achieved an accuracy of 86.00%, meaning it correctly classified most of the test samples. The precision of 73.34% indicates that when the model predicted an individual as belonging

to the >50K income category, 73.34% of those predictions were correct. The recall of 65.24% shows that the model successfully identified 65.24% of all individuals who actually belonged to the >50K category. The F1-score of 69.05% reflects the overall balance between precision and recall, indicating a balanced classification performance.

Confusion Matrix:



The model correctly predicted 6,868 individuals in the <=50K class and 1,524 individuals in the >50K class. It incorrectly classified 554 <=50K individuals as >50K and 812 >50K individuals as <=50K. Overall, the model made more correct predictions than incorrect ones, showing good classification performance.

Pros:

- Achieved highest accuracy among all supervised learning models (86.0%)

- Able to handle a large number of features after One-Hot encoding.
- Can learn complex relationships between the input features.

Drawbacks:

- Difficult to interpret compared to other supervised learning models.
- Performance depends on choices such as number of layers, neurons, epochs, and learning rate.
- Training a neural network is more computationally demanding and time consuming than other simpler models.

2. LOGISTIC REGRESSION

Logistic Regression was used as a classification model to predict the two income categories. The model estimates the probability of an individual belonging to the >50K class and uses this probability to make the final classification.

Evaluation Results:

Logistic Regression Performance Overview

Accuracy: 0.8536585365853658

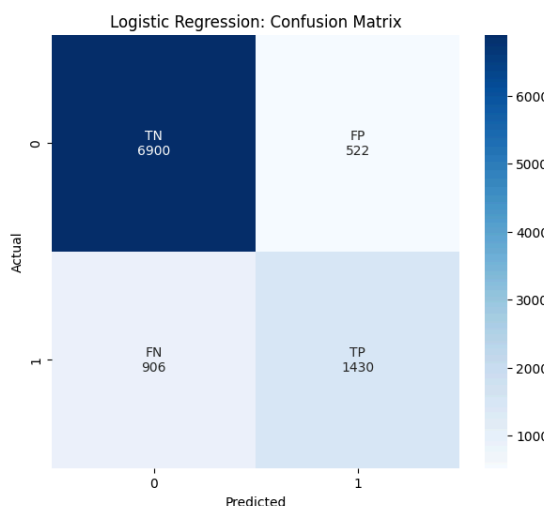
Precision: 0.7325819672131147

Recall: 0.6121575342465754

F1 Score: 0.6669776119402985

Logistic Regression achieved an accuracy of 85.37%, meaning it correctly classified most of the test samples. The precision of 73.26% means that 73.26% of the individuals predicted as >50K actually belonged to that category. The recall of 61.22% means that the model correctly identified 61.22% of all actual >50K individuals. The F1-score of 66.70% shows the balance between precision and recall. Overall, the model performed well, but its recall was lower than the Neural Network's recall.

Confusion Matrix:



The model correctly predicted 6,900 individuals with income $\leq 50K$ and 1,430 individuals with income $> 50K$. It incorrectly classified 522 $\leq 50K$ individuals as $> 50K$ and 906 $> 50K$ individuals as $\leq 50K$. This shows that the model was better at identifying the $\leq 50K$ class and missed a larger number of individuals from the $> 50K$ class.

Pros:

- Simple and fast to train
- Easier to interpret compared to NN.
- Achieved highest accuracy after NN, showing good overall performance

Drawbacks:

- Mainly works with linear relationships, so may not catch complex/non-linear patterns
- Lower recall score than NN, meaning missed identifying $> 50K$ individuals.
- Requires feature scaling for skewed numerical features.

3. DECISION TREE

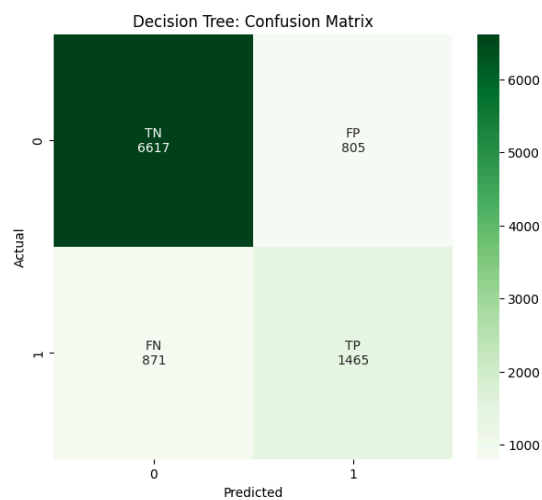
A Decision Tree was used to classify individuals into the two income categories. The model makes predictions by creating a series of decision rules based on the input features.

Evaluation Results:

Decision Tree Performance Overview	
Accuracy:	0.8282434925189588
Precision:	0.6453744493392071
Recall:	0.6271404109589042
F1 Score:	0.6361267911419887

The Decision Tree achieved an accuracy of 82.82%, meaning it correctly classified most of the test samples. The precision of 64.53% means that 64.53% of the individuals predicted as >50K actually belonged to that category. The recall of 62.71% means that the model correctly identified 62.71% of all actual >50K individuals. The F1-score of 63.61% shows the balance between precision and recall. Overall, the Decision Tree performed lower than both the Neural Network and Logistic Regression.

Confusion Matrix:



The model correctly classified 6,617 individuals as <=50K and 1,465 individuals as >50K. It incorrectly classified 805 <=50K individuals as >50K and 871 >50K individuals as <=50K. The model made more errors than the other two models, particularly in predicting the <=50K class as >50K.

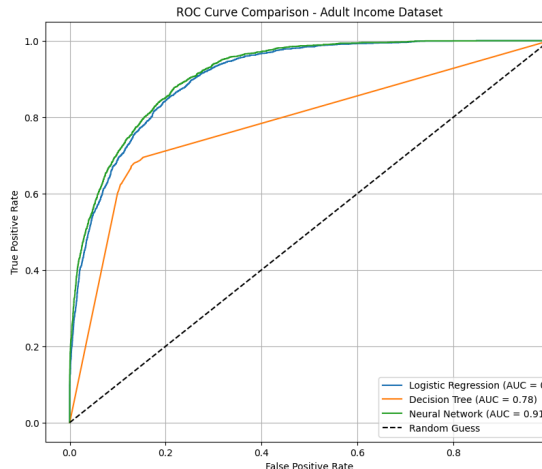
Pros:

- Can capture non-linear relationships between features.
- Easy to understand because it makes decisions using simple rules.
- Does not require feature scaling.

Drawbacks:

- Had the lowest accuracy (82.82%) among the three models.
- Had the lowest precision (64.53%) and F1-score (63.61%).
- Decision Trees can overfit if the tree becomes too complex.

COMPARATIVE ANALYSIS OF MODELS



The ROC curve shows how well each model can distinguish between the $\leq 50K$ and $> 50K$ income classes at different classification thresholds. The dashed diagonal line represents random guessing, so curves farther above this line indicate better performance.

- Logistic Regression $\rightarrow$ AUC = 0.91: The model has a strong ability to distinguish between the two income classes.
- Neural Network $\rightarrow$ AUC = 0.91: It has almost the same overall classification ability as Logistic Regression.
- Decision Tree $\rightarrow$ AUC = 0.78: It performs worse than the other two models, although it is still better than random guessing.

The Neural Network and Logistic Regression performed the best based on AUC, both achieving 0.91. Their curves are also very close to each other and remain well above the

random-guess line. The Decision Tree has a lower AUC of 0.78, showing weaker ability to distinguish between the two income categories.

K-FOLD CROSS VALIDATION

5-fold stratified cross-validation was applied to the training data to check the consistency of each of the model's performance. The training data was divided into five parts, where four parts were used for training and the remaining part was used for validation. This process was repeated five times so that each part was used for validation once. The average score across the five folds was used to assess the overall consistency of the model.

Neural Network

```
Neural Network
-----
Fold Scores: [0.8538, 0.8515, 0.8573, 0.8517, 0.8613]
Mean CV Accuracy: 0.8551
```

Logistic Regression

```
Logistic Regression
-----
Fold scores:
[0.8487255 0.84706033 0.85088394 0.84499103 0.85959518]
Mean CV Accuracy:
0.8502511968784724
```

Decision Tree

```
Decision Tree
-----
Fold scores:
[0.81683105 0.81260407 0.80976172 0.82001025 0.82346913]
Mean CV Accuracy:
0.8165352437836966
```

COMPARE MODEL PERFORMANCE

The table compares the performance of the three supervised models using accuracy, precision, recall, F1-score, and AUC. The Neural Network performed the best overall, achieving the highest accuracy, precision, recall, F1-score, and AUC. Logistic Regression performed slightly lower, while the Decision Tree had the lowest scores across all metrics. Therefore, the Neural Network was selected as the best-performing model for this dataset.

	Model	Accuracy	Precision	Recall	F1 Score	AUC
0	Neural Network	0.859090	0.741579	0.631421	0.682081	0.915
1	Logistic Regression	0.853659	0.732582	0.612158	0.666978	0.907
2	Decision Tree	0.828243	0.645374	0.627140	0.636127	0.782

Best Model Based on Each Score

	Metric	Best Model	Score
0	Accuracy	Neural Network	0.86
1	Precision	Neural Network	0.74
2	Recall	Neural Network	0.63
3	F1 Score	Neural Network	0.68

UNSUPERVISED MODEL TRAINING AND EVALUATION

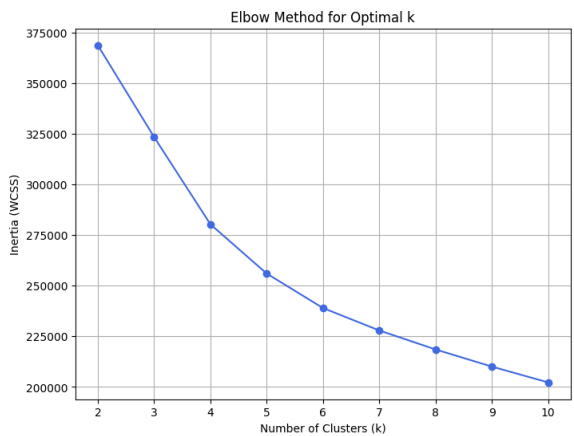
K-MEANS CLUSTERING

After the supervised models, we added K-Means clustering to explore natural groupings in the data independent of the income label.

Data preparation. We reused the same preprocessing pipeline (imputation, scaling, one-hot encoding) and applied it to the full feature set **X**, with no train/test split since clustering is unsupervised. This produced an 88-dimensional feature matrix.

Data prepared for clustering: (48790, 88)

Choosing k. We tested $k = 2$ through 10 using the elbow method, plotting inertia (within-cluster sum of squares) against k



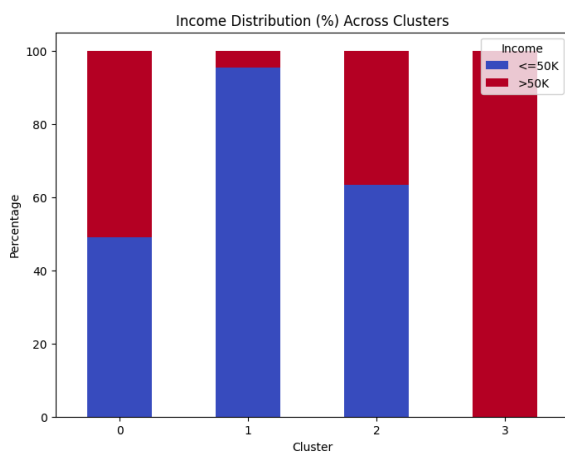
The inertia dropped sharply up to $k=4$ (368,614 $\rightarrow$ 255,996), then flattened

noticeably from $k=4$ onward (the $k=4 \rightarrow 5$ drop was nearly half the size of the $k=3 \rightarrow 4$ drop). This bend marked $k=4$ as the point of diminishing returns, so we set **optimal_k = 4**.

Fitting and profiling. With $k=4$, cluster sizes came out uneven $\rightarrow 25,364 / 20,945 / 2,237 / 244$.

```
... Cluster sizes:
Cluster
2    25364
1    20945
0     2237
3       244
Name: count, dtype: int64
```

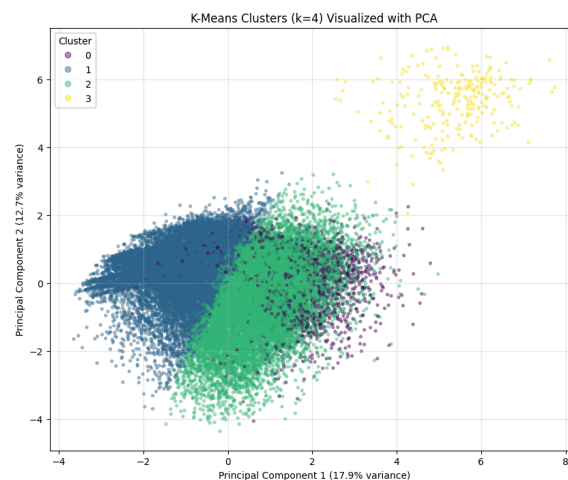
We then cross-tabbed each cluster against the income target and profiled each cluster's average numeric feature values to interpret what each group represented:



- **Cluster 0** ($n=2,237$): ~51% earn >50K, high capital-loss (~\$1,899 avg) $\rightarrow$ likely investors realizing losses.
- **Cluster 1** ($n=20,945$, largest): only ~4.7% earn >50K $\rightarrow$ youngest group (avg age 29), lowest education years (9.7), fewest hours/week (36) $\rightarrow$ early-career/entry-level workers.

- **Cluster 2** ($n=16,047$): ~37% earn >50K, mid-40s age, moderate capital gains (~\$938) $\rightarrow$ general working population.
- **Cluster 3** ($n=244$, smallest): 100% earn >50K, capital-gain pinned at the dataset's max value (99,999), highest education (13 years) and hours/week (50.5) $\rightarrow$ a small, distinct high-income outlier group.

Visualization. We reduced the 88-dimensional feature space to 2D with PCA for a scatter plot, though the first two components only captured ~30.6% of total variance, so the 2D view is an approximation rather than a full picture of cluster separation.



Challenge. Without a second validation metric, k selection relied solely on visually judging the elbow bend, which is more subjective than a metric-driven peak. We mitigated this by comparing the actual magnitude of inertia drops between consecutive k values rather than eyeballing the curve alone.

CONCLUSION

This project set out to predict whether an individual's annual income exceeds \$50K using the Adult Income dataset, combining supervised classification with exploratory unsupervised analysis.

After cleaning the data (removing duplicates, converting “?” placeholders to nulls, dropping redundant/uninformative columns like Education and Final Weight) and building a shared preprocessing pipeline (median/most-frequent imputation, standard scaling, one-hot encoding), we trained and compared three supervised models: a Neural Network, Logistic Regression, and a Decision Tree, validated with stratified 5-fold cross-validation.

The Neural Network performed best overall, leading on accuracy (85.9%), recall (0.670), and F1 score (0.695), while Logistic Regression achieved the highest precision (0.733). The Decision Tree consistently underperformed the other two across every metric, most notably on AUC (0.783 vs. 0.916/0.908), suggesting it overfit to training splits without pruning or depth constraints. Overall, the results indicate the underlying decision boundary between income classes is only mildly non-linear, the Neural Network's edge over Logistic Regression was real but modest, not dramatic.

The K-Means clustering component complemented this by asking a different question: rather than predicting income directly, what natural subgroups exist in the population, and how does income distribute across them without ever using the label to

train on? Using the elbow method, we settled on $k=4$ clusters, which, despite modest separation quality due to the high-dimensional encoded feature space, surfaced interpretable segments ranging from a large, young, low-income-skewed group to a small but distinctly high-income outlier cluster defined by capped capital gains and high education.

Together, the two halves of the project reinforce a consistent story: education level, hours worked, age, and capital gains/losses are the strongest drivers of income in this dataset, showing up both as top features in the supervised models and as the defining traits separating the unsupervised clusters. The main limitations were class imbalance in the target variable, the Decision Tree's tendency to overfit, and the reduced expressiveness of Euclidean-distance clustering once categorical variables were one-hot encoded into a sparse, high-dimensional space, worth flagging as directions for future work (e.g., feature selection before clustering, or trying Gower distance/k-prototypes for mixed data types).